



**SWAMINARAYAN
UNIVERSITY**
THE RESERVOIR OF ALL KNOWLEDGE

Faculty of Engineering (DIPLOMA)

Semester –5TH

Course Title – Basic Of Operating System

Course Code – DE104304

Types of Course -Program Core

Teaching and Examination Scheme –

Hours per Week			Course Credits	Examination Marks				150
L	T	P		Theory Marks		Tutorial /Practical		
				E	M	I	V	
3	0	2	5	60	40	20	30	

Detailed Syllabus –

Unit No.	Content	No. of Hours	% of Weight age
1.	Introduction of Operating System: Fundamental Goals of Operating system , Overview of Operating Systems , Multi programming,Time Sharing , Real Time, Multithreading , Distributed , Operating System services , Case Study ,Linux ,Latest Windows Operating System ,Generations of Operating System	7	20
2.	Process Management and Inter Process Communication: Concept of Process and Threads. Process States and Lifecycle. Process Control Block. Process Scheduling: Types of Schedulers. Scheduling Criteria : CPU utilization, Throughput, Turnaround Time,	8	24

	Waiting Time, Response Time Scheduling Algorithm: First Come First Serve, Shortest Job First, Round Robin. Inter Process Communication Critical Section Semaphore Race condition Mutual Exclusion Deadlock Deadlock characteristics Deadlock Prevention Deadlock Avoidance Deadlock Detection and Recovery		
3.	Memory Management : Logical and physical address map ,Swapping ,Contiguous memory allocation ,Fixed and variable partition ,Internal and External Fragmentation and compaction, Memory relocation and protection mechanism , Allocation techniques – First Fit, Best Fit and Worst Fit , Non Contiguous Memory allocation ,Overview of Paging , Address translation using basic method of paging , Overview of Segmentation	6	22
4.	File Management System: Files System, Files Attributes , File Operations , File Types Directory System , Directory Structures , Protection , Allocation Methods – Contiguous, Linked Secondary Storage Structure ,Disk Structure,Disk Scheduling Algorithm – SCAN, CSCAN	5	18
5.	Digital Promotions Linux Introduction , Basic architecture of Unix/ Linux , Introduction to shell and commands , Commands: pwd, cd, mkdir, rmdir, ls, cat, cp, rm, mv, wc, split, cmp, comm, diff, head, tail, grep, sort, Editing files with “vi”, “vim”, “gedit”, “gcc” Linux Basic shell scripts, Read using command line argument , the logical operators , Evaluate expression using test and ,Branching : if, case , Basic of Looping : while, for	4	16
	Total	30	100

Course Outcome:

After Completion of the Course, Student will able to:

No	Course Outcomes	RBT level
1	Understand the basic concepts of Operating System.	Understand
2	Illustrate various aspects of process scheduling and deadlock management	Understand
3	Understand memory management concept.	Understand
4	Understand various file management and file allocation techniques.	Understand
5	Develop basic shell script using operators, control and loop statements.	Apply

References/Suggested Learning Resources:

(a) Books:

1. Operating System Concepts, Abraham Silberschatz, Peter B. Galvin, Greg Gagne, 9 th Edition, WILEY
2. Modern Operating Systems Andrew S. Tanenbaum & Herbert Bos 5 th Edition, Pearson
3. Operating Systems: Internals and Design Principles William Stallings 7 th Edition, Pearson
4. Linux Command Line and Shell Scripting Bible. Richard Blum & Christine Bresnahan 3 rd Edition, WILEY
5. Operating Systems. Dhananjay M. Dhamdhare MGH, 2017 ISBN 978-0-07-295769-3
6. Principles of Operating Systems. Naresh Chauhan, Oxford University Press, New Delhi, 2014, ISBN : 9780198082873
7. Linux –Application And Administration Ashok Kumar Harnal TMH, 2009, ISBN - 1283188996, 781283188999

(b) Open source software and website:

1. https://www.tutorialspoint.com/operating_system/index.htm
2. <https://www.tpointtech.com/operating-system>
3. <https://www.studytonight.com/operating-system/>
4. <https://linuxcommand.org/>
5. Linux basics: www.freeos.com/guides/lsst
6. Linux basics: www.linuxcommand.org/writing_asell_scripts.php
7. Linux basics: <https://nptel.ac.in/courses/117106113>
8. Linux basics : https://onlinecourses.swayam2.ac.in/aic20_sp24/preview

Suggested Course Practical List:

SR NO.	PRACTICAL
1	Compare following Operating System:

	a) Compare multiprogramming and multiprocessing Operating System. b) Compare multiprocessing and multitasking Operating System
2	Implement CPU scheduling algorithm FCFS using Python programming for a given table and calculate average waiting time and average turnaround time. Do a comparative analysis of their landing pages
3	Test and execute Linux process commands for following task: a) Execute top command to display the most CPU- and memoryintensive processes running on your system. b) Execute ps command to display information about currently running process on your system. c) Execute kill command to stop various running process on your system.
4	Test and execute Linux file and directory commands to perform following task: a) List the directory content of your system. b) Create “NEP2020” directory. c) Change your current directory to “NEP2020” directory. d) Display the full path of your current working directory. e) Rename “NEP2020” directory with “NewEdu”. f) Delete “NewEdu” directory. g) Write a Linux command to clear your screen. h) Display the list of 20 previously executed command.
5	Test and execute calendar and date command in Linux to perform following task: a) Display the calendar for the month March 1927. b) Display the calendar for the year 2050. c) Display the date and time of your system. d) Display the date of your system in mm/dd/yyyy format for e.g. 07/14/2030. e) Sleep your system for 20 second.
6	Test and execute the Linux user command and networking command to perform following task: a) Display the user id of the currently logged-in user of your system. b) Display the host name of your system. c) Execute the Linux commands: who, w and last. d) Display the IP address of your system.
7	Test and execute Linux file and file editing commands to perform following task: a) Write a shell script to perform following task: 1) Read “NEP2020” from the command prompt and create directory “NEP2020”. 2) Read “NewEduPolicy” from the command prompt and rename “NEP2020” directory to “NewEduPolicy”. 3) Display the full path of your current working directory. b) Write a shell script to perform following task: 1) Read “Chandrayan2.txt” from the command prompt and create a blank file with the name “Chandrayan2.txt” 2) Read “Chandrayan2.txt” from the command prompt to open & write at least 10 lines in the “Chanrayan2.txt” file.

	3) Read “Chandrayan2.txt” from the command prompt to display the content of “Chanrayan2.txt” file. c) Write a shell script to read two different file names from the command prompt and copy the content of first file into the second file. d) Write a shell script to read “Chandrayan2.txt” file name and “ISRO” word from the command prompt and search “ISRO” word from “Chandrayan2.txt” file.
8	Test and execute wc command: a) Write a shell script to accept the string “diploma in information technology” from the command prompt in the lower case letter and convert it into the upper case letter. b) Write a shell script to read the “Chandrayan2.txt” file from the command prompt and display the numbers of characters, words & lines of a given input file “Chandrayan2.txt”.
9	Understand and Apply Arithmetic Operators: Write a shell script to perform arithmetic operations: a. Write a shell script to read two numbers from the command prompt and perform addition, subtraction, multiplication, division and modulus operation. b. Write a shell script to read the marks of five courses out of 100 from the command prompt and calculate the total marks and percentage (%) of the marks. c. Write a shell script to read radius (R) of circle in cm from the command prompt and calculate the area (A) of circle.
10	Understand and apply control statements: Write a shell script to perform following operations: a) Write a shell script to read three numbers from the command prompt and display maximum number among three numbers. b) Write a shell script to read the marks of five courses out of 100 from the command prompt and calculate: 1) The total marks. 2) The percentage (%) of the marks. 3) The grade: First class, Second class, Third class based on percentage (%). 4) The result pass or fail based on the marks of five courses. c) Write a shell script to read N numbers from the command prompt and display the sum and average of N numbers. d) Create a shell script to read 6 digit number from the command prompt and display the reverse digits of a given 6-digit number. (For e.g. if the no. is 574291 then answer is 192475).

List of Laboratory/Learning Resources Required:=

1. Computer system with Linux Operating System.
2. Python IDEs and Code Editors.

List Suggested Project List:

- a) Simulate SJF, Round Robin, and Priority Scheduling using a Python program.
- b) Compare average waiting and turnaround time.
- c) Write a shell script to display top 5 memory-consuming processes.
- d) Script to back up a directory.
- e) Login welcome script using echo and system variables.
- F) Create users and groups using adduser, groupadd.
- g) Assign permissions and group memberships
- h) Use passwd to manage passwords.